EduPilot AI

Your AI Co-Pilot for Teaching & Adaptive Learning

Comprehensive Judge Presentation & Live Demonstration Blueprint

Project Summary & High-Level Value Proposition

Domain: Agentic AI and Autonomous Pedagogical Workflows

Problem: One-size-fits-all education leaves students struggling with dense PDFs; teachers spend 60%+ time on repetitive tasks.

Solution: Transforms static course PDFs into personalized, multi-tier AI explanations, auto-generated assessments, and real-time analytics.

Key USP: Grounded RAG PDF Simplification with dynamic Beginner / Intermediate / Advanced explanation adaptation.

1. Problem Statement & The EduPilot AI Solution

The Classroom Reality (The Problem):

In traditional education systems, teachers provide textbooks, PDFs, and notes designed for the "average" student. However, students learn at vastly different speeds, with varying foundational knowledge and learning preferences. When students study alone, complex academic language in PDFs causes frustration, doubts remain unresolved, and struggling students quietly fall behind. Meanwhile, educators spend over 60% of their working hours manually preparing materials, formatting quizzes, and grading rather than mentoring.

The EduPilot AI Solution:

EduPilot AI creates a complete, closed-loop educational ecosystem bridging teachers and students:

- **Teacher Co-Pilot:** Rapidly ingests raw PDFs, organizes modular courses, and drafts diagnostic assessments using AI.
- **Student AI Tutor:** Answers doubts 24/7 with RAG-grounded citations and simplifies complex topics to three selectable comprehension tiers.
- **Closed-Loop Analytics:** Continuously tracks student quiz scores and alerts teachers to exact classroom knowledge gaps.

2. Complete 5-Feature Live Demonstration Walkthrough

Present these 5 core features in sequence to give the judges a seamless, compelling demonstration:

Feature 1: Teacher Uploads Courses & Educational Materials

- **What it is:** The teacher creates courses (e.g., "Introduction to AI & Machine Learning") and uploads syllabus PDFs and notes.
- **How it works:** The system parses PDFs using unpdf, extracts text, cleans formatting, and generates semantic vector chunks.
- **What to show in Demo:** Open the Teacher Studio (teacher@edupilot.ai), navigate to "Courses", and highlight the ready-to-use course materials.

Feature 2: Students Access Courses & Personalized Dashboard

- **What it is:** Students log into their dedicated portal to access enrolled courses, view syllabus objectives, and track mastery.
- **How it works:** Role-based authentication ensures students see their individualized learning path and verified course modules.
- **What to show in Demo:** Open the Student Dashboard (student.demo@edupilot.ai), showing active enrollments and progress metrics.

Feature 3: Teacher Generates & Assigns Assessments via AI

- **What it is:** Teacher Co-Pilot automatically drafts diagnostic multiple-choice questions grounded in course concepts.
- **How it works:** Powered by Agnes/Gemini AI with strict Zod schema validation; ensures 4 options, clear pedagogical rationales, and difficulty levels.
- **What to show in Demo:** Click "Generate Questions with AI", watch 5 high-quality questions appear in seconds, and show teacher approval controls.

Feature 4: Student AI Tutor with RAG PDF Simplification (The Star Feature)

- **What it is:** An interactive full-screen AI Tutor that breaks down complex PDF topics into student-friendly explanations.
- **RAG Grounding:** Retrieves exact excerpts from the uploaded PDF so answers are 100% verified with clickable citations (No Hallucinations).
- **3-Tier Complexity Toggling:** Students click Beginner (simple analogies), Intermediate (core mechanics), or Advanced (mathematical rigor).
- **What to show in Demo:** Ask "What is gradient descent?", toggle between Beginner and Advanced, and point out the grounded PDF citations.

Feature 5: Teacher Diagnostic Analytics & Quiz Gap Detection

- **What it is:** Real-time classroom heatmaps showing mastery scores across individual concepts.
- **How it works:** Quiz submissions automatically update student profiles, highlighting struggling topics (e.g. Neural Networks).
- **What to show in Demo:** Show the Teacher Analytics view displaying concept mastery radars and struggle alerts.

3. Judge Presentation Script (Word-for-Word Presenter Guide)

Part 1: The 45-Second Opening Hook

"Respected judges, every student in a classroom understands at a different pace, yet traditional learning gives everyone the exact same textbook PDF. When students get stuck studying alone, they lose confidence. At the same time, teachers spend over 60% of their time formatting quizzes and grading instead of teaching.

We built EduPilot AI - an autonomous co-pilot that turns static teacher PDFs into an adaptive learning experience. Let us show you how it works in real-time."

Part 2: The 3-Minute Live Demo Execution

- **Step 1 (Teacher):** "Here on the Teacher Portal, I have uploaded our course syllabus. Instead of spending hours writing test questions, I click 'Generate with AI', and our Co-Pilot creates 5 curriculum-aligned diagnostic questions in seconds."
- **Step 2 (Student):** "Now, logging in as a student, I have access to this course. When I encounter a difficult concept, I open our AI Tutor workspace."
- **Step 3 (RAG & Tiers):** "Notice two critical innovations: First, every answer cites the exact page from the teacher's PDF. Second, I can toggle from Beginner - which uses simple everyday analogies - to Advanced, which gives me rigorous mathematical formulas."
- **Step 4 (Analytics):** "Finally, as students complete diagnostic quizzes, the teacher dashboard instantly surfaces class-wide struggle points, allowing the teacher to take immediate corrective action."

Part 3: The Closing Statement

"EduPilot AI is not just a chatbot - it is a complete, continuous learning loop that saves teachers hours of repetitive work while ensuring every student receives personalized, grounded guidance. Thank you, and we welcome your questions!"

4. Live Demo Credentials & System Cheat Sheet

Quick-Access Demo Credentials (Pre-Seeded & Ready)

Teacher Account: teacher@edupilot.ai | Password: TeacherPass123! (Preloaded with 11 courses)
Student Account: student.demo@edupilot.ai | Password: Password123! (Enrolled & Ready)
Local Demo URL: http://localhost:3000 (Run "npm run dev")
Frontend Stack: Next.js 16 (App Router), React 19, Tailwind CSS v4, Lucide Icons
Backend & Data: Firebase Auth & Cloud Firestore (User & Course Data), Supabase (Storage)
AI Engine: Agnes AI / Gemini 3.0 Flash with Structured Output Schemas & RAG Vector Retrieval